

Project Design Phase-II

Technology Stack (Architecture & Stack)

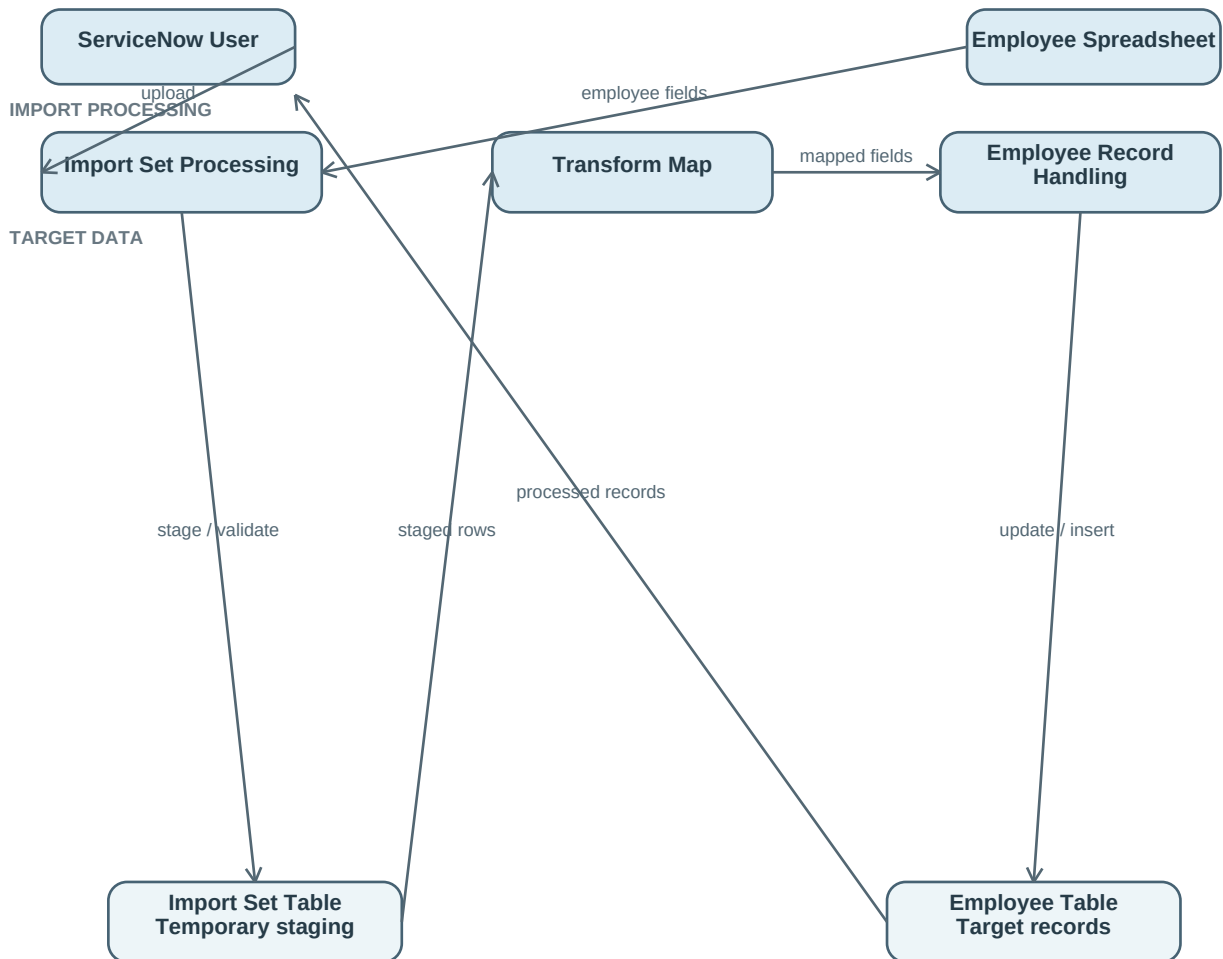
Date	1 October 2026
Team ID	SWTID-2026-3453
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Technical Architecture:

The deliverable includes the technical architecture and the technology information required for the application. For this project, the workflow is implemented within the ServiceNow platform using its web interface, Import Sets, Import Set Table, Transform Map, and target Employee table.

Technical Architecture – ServiceNow Employee Spreadsheet Import

USER / INPUT



Platform-managed ServiceNow infrastructure; no separate application server or database is required for the stated workflow.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	How the user interacts with the import workflow.	ServiceNow Web UI
2	Application Logic-1	Receives and stages employee spreadsheet records for temporary processing and validation.	ServiceNow Import Sets / Import Set Table
3	Application Logic-2	Maps source spreadsheet fields to corresponding target Employee fields.	ServiceNow Transform Maps

S.No	Component	Description	Technology
4	Application Logic-3	Identifies existing employees and handles update or insertion using Employee ID.	Transform Map field mapping and Employee ID matching
5	Database	Stores the staged import records and target employee records within the ServiceNow platform.	ServiceNow platform-managed data storage; Import Set Table and Employee table
6	Cloud Database	No separate cloud database service is required by the stated solution.	Not separately used; ServiceNow platform storage
7	File Storage	Employee spreadsheet acts as the input data source for the import process.	Spreadsheet file supplied to ServiceNow import workflow
8	External API-1	No external API is required for the stated employee spreadsheet import workflow.	Not used
9	External API-2	No second external API is required for the stated workflow.	Not used
10	Machine Learning Model	No machine learning model is required for spreadsheet import, field mapping, or Employee ID matching.	Not used
11	Infrastructure (Server / Cloud)	Application runs on the ServiceNow platform rather than on a separately managed application server.	ServiceNow cloud platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No separate open-source framework is used in the stated ServiceNow import workflow.	Not separately used
2	Security Implementations	Access to the ServiceNow application and import workflow is governed by the platform's configured user access and permissions.	ServiceNow access controls / role-based permissions
3	Scalable Architecture	The workflow uses ServiceNow platform services for staging, transformation, and target record processing instead of a separately managed multi-server application.	ServiceNow Import Sets and Transform Maps
4	Availability	The solution is available when the ServiceNow instance and required import functionality are available.	ServiceNow platform
5	Performance	Performance is supported by a staged import flow and direct field mapping without additional application-layer processing.	Import Set staging and Transform Map processing

References:

C4 Model – <https://c4model.com/>

IBM Cloud Architecture – <https://www.ibm.com/cloud/architecture>

AWS Architecture – <https://aws.amazon.com/architecture>